

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30 % of CIA	Total Marks:	20 (Answer ANY ONE)
CO3 Statement:	<i>Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)</i>		
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Section / Batch:	SLOT D / 2024	Date:	

Code of Conduct

I KAVIYA M 192421050 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

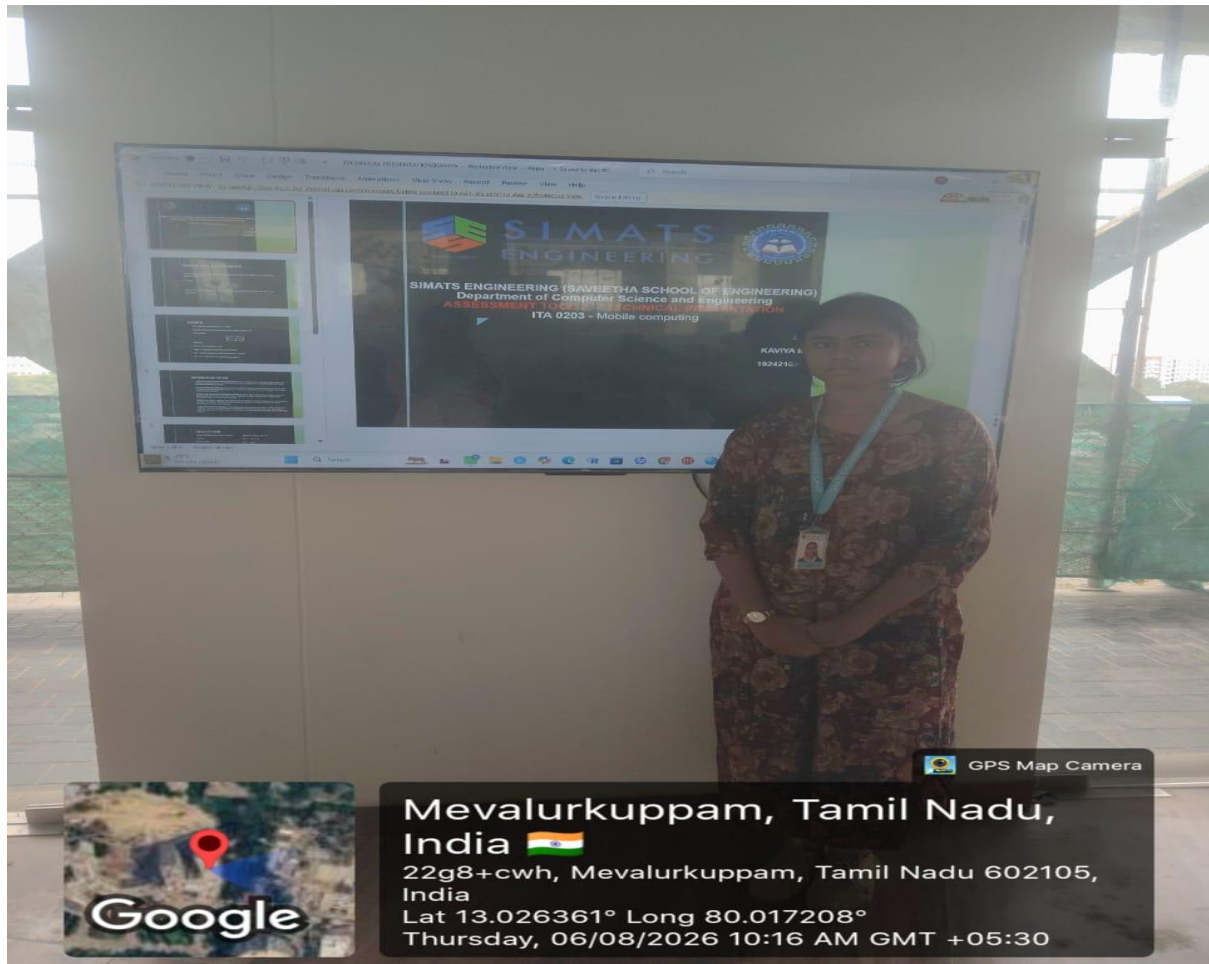
Signature of the Student: _____

Instructions

- Duration: 10 minutes

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. A CDMA system has a processing gain of 256 and requires a linear SIR of 6. Using $SIR = PG/(N-1)$, derive the maximum number of simultaneous users the system can support, showing the algebraic rearrangement step by step. Discuss why this relationship makes CDMA capacity "soft" (interference-limited) rather than hard-limited like FDMA/TDMA channel counts, and what this means operationally.





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ITA 0203 - Mobile computing

BY

KAVIYA M

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PROBLEM STATEMENT

A CDMA system has a **Processing Gain (PG) = 256** and requires a **linear SIR = 6**.

Using ,

$$SIR = \frac{PG}{N - 1}$$

derive the maximum number of simultaneous users the system can support.

GIVEN

- Processing Gain (PG) = 256
- Linear Signal-to-Interference Ratio (SIR) = 6
- Formula:

$$SIR = \frac{PG}{N - 1}$$

- Where
- PG = Processing Gain
- SIR = Signal-to-Interference Ratio
- N = Total number of simultaneous users
- N - 1 = Number of interfering users

INTRODUCTION

- **CDMA (Code Division Multiple Access)** is a mobile communication technology that allows multiple users to communicate simultaneously on the **same frequency** using **unique spreading codes**.
- **Processing Gain (PG)** is the amount by which a signal is spread before transmission. A **higher processing gain** improves resistance to interference and allows more users to share the network.
- **Linear SIR (Signal-to-Interference Ratio)** is the ratio of the desired signal power to the interference power, expressed as a **simple number** (not in dB). A higher SIR indicates better communication quality.
- **CDMA has Soft Capacity** because all users share the same frequency. As the number of users increases, interference increases gradually, reducing the SIR instead of blocking new users immediately.
- **Objective of this Problem:** Using the given Processing Gain (PG = 256) and Linear SIR (6), calculate the **maximum number of simultaneous users** that the CDMA system can support and explain why CDMA is an **interference-limited (soft capacity)** system.

SOLUTION

- **Substitute the Given Values**
- Given:
- **PG = 256**
- **SIR = 6**
- Substitute into the formula:
- $6 = \frac{256}{N-1}$
- **Step 3: Cross Multiply**
- $6(N - 1) = 256$
- **Step 4: Expand the Equation**
- $6N - 6 = 256$
- **Step 5: Solve for N**
- $6N = 256 + 6$
- $6N = 262$
- $N = \frac{262}{6}$
- $N = 43.67$
- $N = 43$

VERIFICATION

- **Verification for 43 Users**
- Number of interfering users:
- $N - 1 = 43 - 1 = 42$
- Substitute into the formula:
- $SIR = \frac{256}{42}$
- $SIR = 6.095$
- Since,
- $6.095 > 6$

THEREFORE , The required SIR is satisfied.